Mini Project Report
on
ML Based Early Disease Risk Prediction

Group 9: Tankasali Bharath, Vishnu Gupta, Gokul Raj, Debaprasada Dash

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Github: Project Link

Abstract

The rapid rise in lifestyle-related diseases has become a major global health concern, necessitating the development of intelligent and scalable healthcare solutions. Early detection and continuous monitoring of chronic health conditions can significantly reduce complications, mortality rates, and healthcare costs. With the increasing availability of digital health data and advances in machine learning, predictive healthcare systems have emerged as a powerful tool to assist medical professionals in decision-making and preventive care.

This project focuses on the design and development of an intelligent disease prediction system that leverages machine learning techniques to assess health risks based on patient data. The system analyzes multiple clinical parameters such as age, physiological measurements, and lifestyle-related factors to predict the likelihood of major chronic diseases. Various supervised learning algorithms are employed to learn patterns from historical medical datasets, enabling accurate classification and risk assessment. Data preprocessing techniques, including handling missing values, normalization, and feature selection, are applied to improve model performance and reliability.

The proposed system aims to provide a unified platform capable of supporting multiple disease prediction models within a single framework. This modular and extensible architecture ensures scalability and adaptability for future enhancements. The results demonstrate that machine learning-based disease prediction can serve as an effective decision-support tool, offering timely insights for preventive healthcare. By assisting in early diagnosis and promoting proactive medical intervention, the system contributes to improving healthcare outcomes and supports the transition toward data-driven and personalized healthcare solutions.

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1 Introduction

The healthcare sector is undergoing a rapid transformation driven by the growing availability of digital medical data and advancements in computational intelligence. Lifestyle-related diseases such as cardiovascular disorders, metabolic conditions, and obesity-related complications have significantly increased due to sedentary habits, unhealthy diets, and stress. These diseases often remain undetected during early stages, leading to severe health complications and increased mortality rates. Early diagnosis and preventive care are therefore critical in improving patient outcomes and reducing the burden on healthcare systems.

Machine learning has emerged as a promising approach in the healthcare domain due to its ability to analyze large volumes of data and identify hidden patterns that may not be evident through traditional diagnostic methods. By leveraging historical medical records and patient health parameters, machine learning models can assist in predicting disease risks with considerable accuracy. Such predictive systems can serve as valuable decision-support tools for medical professionals, enabling timely intervention and personalized treatment strategies.

1.1 Motivation

The primary motivation behind this project is the increasing prevalence of chronic lifestyle-related diseases and the limitations of conventional diagnostic approaches. Traditional healthcare systems often rely on reactive treatment methods rather than proactive prevention, which can result in delayed diagnosis and higher medical costs. Additionally, limited access to healthcare professionals in rural and underdeveloped regions further emphasizes the need for automated and intelligent diagnostic support systems.

With the availability of publicly accessible medical datasets and advancements in machine learning algorithms, there is a strong opportunity to develop reliable disease prediction models. The motivation of this project is to utilize these technological advancements to build an intelligent system capable of predicting disease risks at an early stage, thereby supporting preventive healthcare and improving overall quality of life.

1.2 Problem Statement

Despite significant progress in medical technology, early detection of chronic diseases remains a challenge due to the complexity of medical data and the dependency on manual analysis. Many existing systems focus on the prediction of a single disease, limiting their practical applicability in real-world healthcare scenarios where multiple conditions often coexist. Moreover, inconsistent data quality, missing values, and lack of integrated platforms further reduce the effectiveness of traditional diagnostic methods.

The problem addressed in this project is the development of a unified and efficient disease prediction system that can accurately analyze patient health data and assess the risk of multiple chronic diseases. The system aims to overcome data-related challenges and provide reliable predictions using machine learning techniques, thereby assisting healthcare professionals and individuals in making informed medical decisions.

1.3 Objectives

The key objectives of this project are as follows:

- To develop a machine learning–based system for early disease risk prediction.
- To preprocess and analyze healthcare datasets for improved prediction accuracy.
- To implement and compare supervised learning models for disease risk assessment.
- To design a unified framework supporting prediction of multiple chronic diseases.
- To promote preventive healthcare through data-driven decision support.

2 State-of-the-Art Models

Over the years, various statistical and machine learning techniques have been applied to healthcare risk prediction. These models differ in complexity, interpretability, and predictive performance. A structured overview of commonly used approaches is presented below.

2.1 Traditional Statistical Models

Traditional clinical prediction systems primarily rely on statistical approaches due to their simplicity and interpretability.

- **Logistic Regression:** Widely used for binary disease classification problems. It provides probabilistic outputs and allows interpretation of feature coefficients.
- **Linear Models:** Used when relationships between predictors and target variables are assumed to be linear.

Although statistically robust, these models may struggle to capture complex non-linear relationships present in healthcare data.

2.2 Tree-Based Machine Learning Models

Decision tree–based models gained popularity due to their ability to model non-linear feature interactions.

- **Decision Trees:** Provide rule-based classification and are highly interpretable.
- **Random Forest:** An ensemble of decision trees that improves generalization and reduces overfitting.
- **Gradient Boosting (XGBoost, LightGBM):** Sequentially improves weak learners and achieves high predictive accuracy in structured datasets.

These models are particularly effective for tabular healthcare datasets containing mixed categorical and numerical features.

2.3 Support Vector Machines and Kernel Methods

Support Vector Machines (SVM) are effective in high-dimensional feature spaces and can model complex boundaries using kernel functions. However, they may require careful parameter tuning and are less interpretable compared to tree-based methods.

2.4 Neural Network-Based Approaches

Artificial Neural Networks (ANNs) and deep learning models have been explored for disease prediction, especially in large-scale datasets.

- Capable of capturing complex feature interactions.
- Require large datasets and computational resources.
- Often lack transparency in decision-making.

While powerful, deep learning models may not always be suitable for structured healthcare datasets with moderate sample sizes.

2.5 Comparative Overview of Models

Table 1: Comparison of State-of-the-Art Models in Disease Prediction

Model Type	**Interpretability**	**Non-linearity Handling**	**Accuracy Potential**
Logistic Regression	High	Limited	Moderate
Decision Tree	High	Moderate	Moderate
Random Forest	Medium	Strong	High
Gradient Boosting	Medium	Strong	Very High
Neural Networks	Low	Very Strong	Very High

2.6 Relevance to the Proposed Work

Considering the structured nature of the dataset and the importance of interpretability in healthcare applications, this project adopts a hybrid modeling strategy. Logistic Regression is used as a baseline model for transparency, while ensemble techniques such as Random Forest and Gradient Boosting are employed to improve predictive performance. Model comparison is performed using ROC-AUC and other classification metrics to select the best-performing algorithm for disease risk prediction.

3 Research Gaps and Motivation

Although machine learning has been widely applied in healthcare risk prediction, several limitations still exist in current state-of-the-art approaches. A detailed analysis of existing literature reveals gaps in model scope, interpretability, integration, and preventive focus.

3.1 Identified Research Gaps

- **Single-Disease Focus:** Most existing studies concentrate on predicting only one specific disease, limiting their applicability in real-world healthcare scenarios where multiple chronic conditions often coexist.
- **Limited Lifestyle Integration:** Several models rely heavily on clinical or laboratory data while underutilizing behavioral and lifestyle factors that significantly influence chronic disease development.
- **Lack of Interpretability:** Advanced models such as deep neural networks often achieve high accuracy but fail to provide transparent explanations, which is critical in healthcare decision-making.
- **Data Fragmentation:** Many systems are developed for isolated datasets without a unified framework capable of integrating multiple prediction models.
- **Reactive Rather Than Preventive Approach:** Existing healthcare systems primarily emphasize diagnosis after symptom onset rather than early risk awareness and preventive intervention.

3.2 Comparative Gap Analysis

Table 2: Comparison of Existing Approaches and Proposed System

Criteria	**Existing Models**	**Proposed System**
Multi-Disease Support	Limited	Yes
Lifestyle-Based Features	Partial	Extensive
Model Interpretability	Often Low	Emphasized
Unified Framework	Rare	Integrated
Preventive Risk Focus	Moderate	Strong

3.3 Motivation of the Proposed Work

The identified research gaps highlight the need for a unified, interpretable, and preventive disease prediction framework. This project aims to bridge these gaps by developing a

multi-disease risk prediction system that integrates demographic, clinical, and lifestyle features within a single modular architecture.

Unlike traditional approaches that emphasize post-diagnosis prediction, the proposed system focuses on early risk estimation and transparent model explanations. By combining classical statistical techniques with modern ensemble learning methods, the system balances prediction accuracy with interpretability. This approach ensures practical usability in healthcare environments while promoting preventive medical awareness.

4 Dataset Analysis

4.1 Dataset Description

The dataset used in this study consists of publicly available healthcare records containing demographic attributes, lifestyle behaviors, and medical indicators relevant to chronic disease prediction. The dataset includes variables such as age, body mass index (BMI), physical activity, smoking status, alcohol consumption, stress level, sleep duration, blood pressure indicators, and medical history attributes.

The dataset is structured and tabular in nature, making it suitable for supervised machine learning models. The target variable represents disease presence or absence, allowing binary classification–based risk prediction.

The dataset is publicly available at: `https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset`

4.2 Feature Categorization

For better analytical clarity, the features were categorized into logical groups as shown in Table 3.

Table 3: Feature Categorization

Category	**Example Features**
Demographic	Age, Gender
Lifestyle	Physical Activity, Smoking, Alcohol Consumption
Physiological	BMI, Blood Pressure, Cholesterol
Behavioral Indicators	Sleep Duration, Stress Level
Medical History	Previous Conditions, Family History

4.3 Data Preprocessing and Cleaning

The dataset underwent several preprocessing steps before model training:

- Missing values were handled using appropriate imputation strategies.
- Categorical variables were encoded using label or one-hot encoding.
- Numerical features were scaled to ensure uniform model learning.
- Outliers were examined using statistical thresholds.
- The dataset was split into training and testing subsets for evaluation.

4.4 Class Distribution Analysis

Initial exploratory analysis revealed class imbalance in certain disease categories. To ensure fair model training, appropriate strategies such as stratified splitting were used during train-test division. Evaluation metrics such as ROC-AUC were preferred over accuracy to better assess model performance under imbalance conditions.

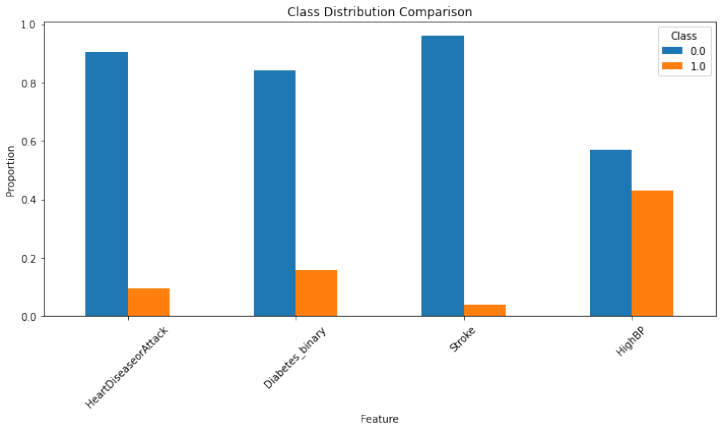

Figure 1: Class Distribution of Target Variable

4.5 Age Distribution

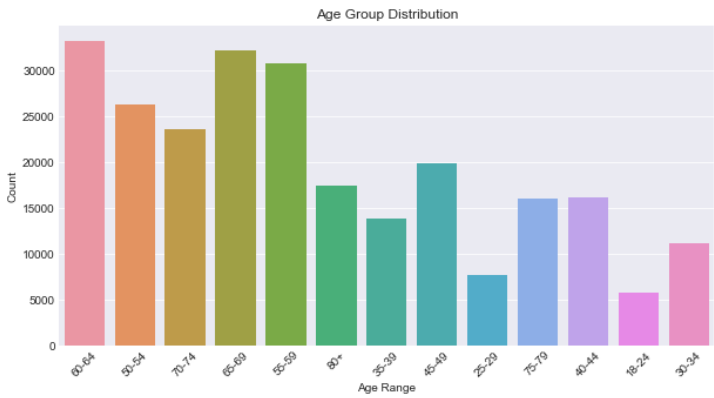

Figure 2: Age Group Distribution

4.6 Correlation and Feature Importance Analysis

A correlation matrix was generated to examine relationships between input variables and identify redundant features. Strong correlations were observed between BMI and lifestyle indicators, as well as between age and physiological risk factors.

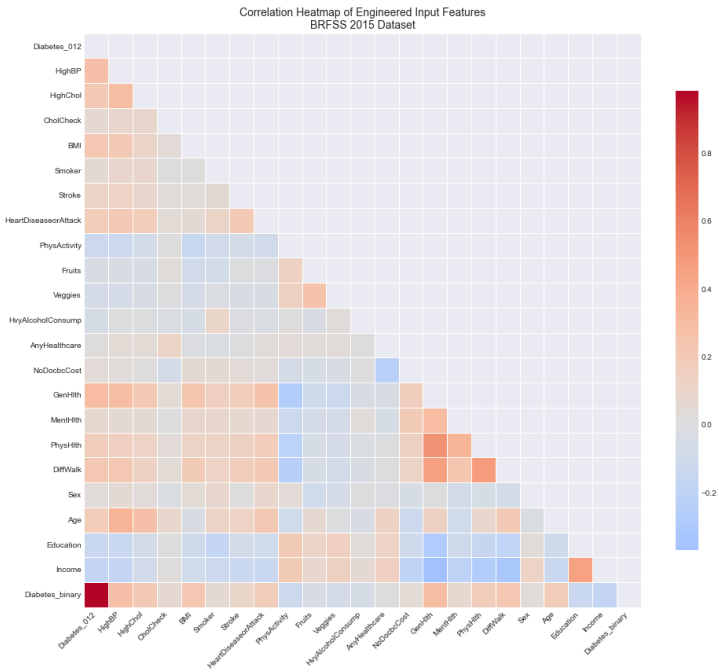

Figure 3: Correlation Heatmap of Input Features

Feature importance analysis was conducted using ensemble models such as Random Forest and Gradient Boosting. The analysis highlighted age, BMI, RiskScore and PreventiveCareIndex as dominant predictors across multiple disease categories.

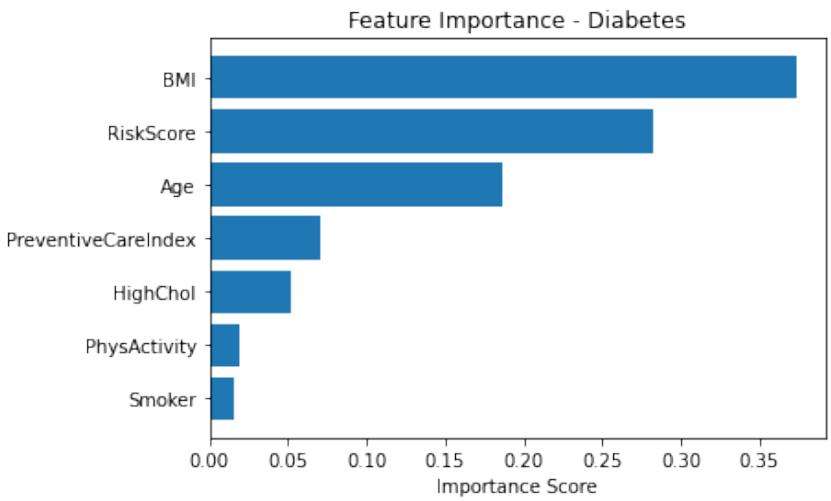

Figure 4: Diabetes Feature Importance Analysis

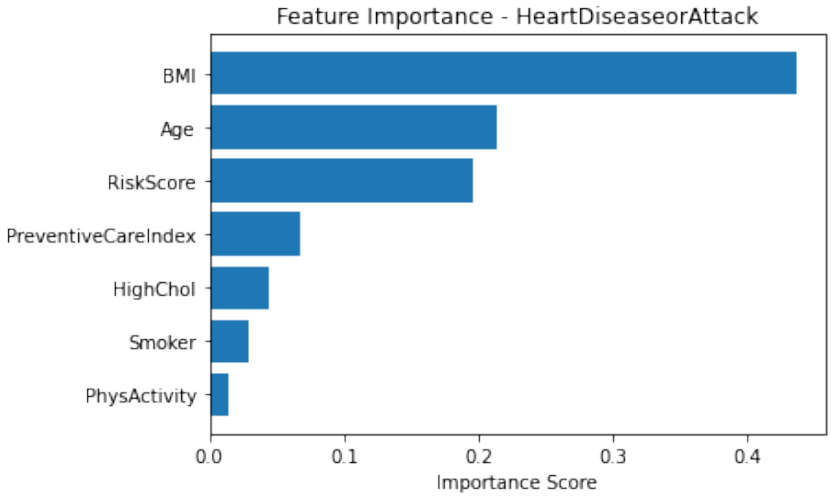

Figure 5: Heart Disease Feature Importance Analysis

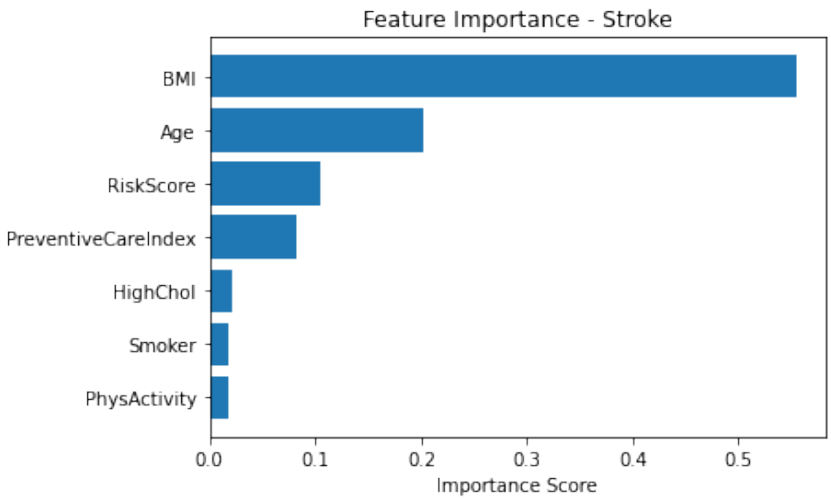

Figure 6: Stroke Feature Importance Analysis

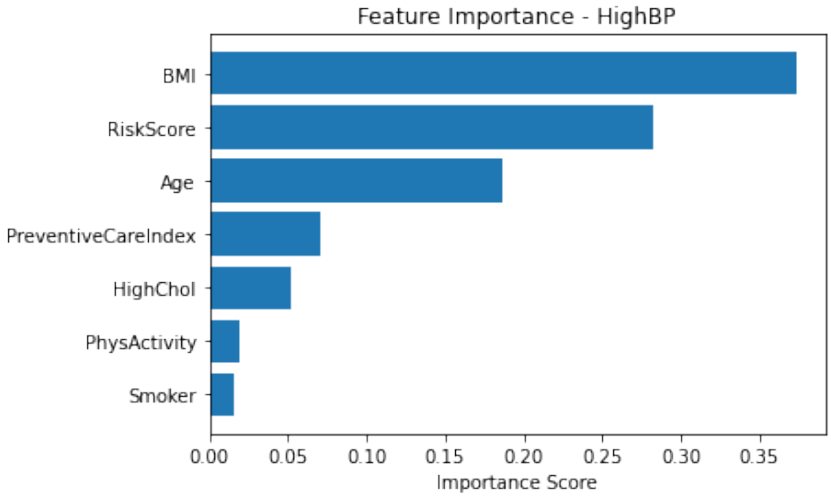

Figure 7: Hypertension Feature Importance Analysis

4.7 Train-Test Split Strategy

The dataset was divided into training and testing subsets to evaluate generalization performance. Stratified sampling was used to preserve class proportions. Probability-based predictions were generated using trained models to compute ROC-AUC scores and evaluate discriminatory power.

4.8 Summary of Dataset Insights

The dataset analysis revealed that lifestyle and physiological features significantly contribute to disease risk prediction. Feature engineering and careful preprocessing improved model stability and predictive capability. The analysis confirmed that structured healthcare datasets, when properly cleaned and processed, can effectively support multi-disease risk prediction within a unified machine learning framework.

5 Methodology and Approach

The proposed system follows a structured machine learning pipeline to perform multi-disease risk prediction using healthcare data. The methodology integrates data preprocessing, feature engineering, class imbalance handling, model training, evaluation, and risk scoring.

5.1 Problem Formulation

The problem is formulated as a **multi-label classification task**, where a single input instance can be associated with multiple disease outcomes simultaneously. The feature matrix is represented as:

$$X = \{x_1, x_2, ..., x_n\}$$

and the target vector is defined as:

$$Y = \{y_1, y_2, y_3, y_4\}$$

where each target corresponds to a specific disease condition.

5.2 Data Preprocessing

The dataset is preprocessed to ensure quality and consistency before model training:

- Missing values are handled using appropriate imputation techniques.
- Categorical variables are encoded into numerical format.
- Numerical features are normalized using StandardScaler.
- The dataset is split into training (80%) and testing (20%) subsets.

5.3 Feature Engineering

Additional features are derived to enhance predictive capability:

- **PreventiveCareIndex:** Represents overall health awareness and preventive behavior.
- **RiskScore:** Aggregated indicator combining multiple lifestyle risk factors.

These engineered features help capture complex interactions between lifestyle variables.

5.4 Handling Class Imbalance

The dataset exhibits significant class imbalance, particularly for diseases such as stroke and heart disease. To address this issue:

- Class weighting techniques were initially applied.
- Synthetic Minority Over-sampling Technique (SMOTE) was used to generate synthetic samples for minority classes.

SMOTE was applied separately for each disease to improve minority class representation during training.

5.5 Model Development

Multiple machine learning models were implemented and evaluated:

- Logistic Regression (Baseline Model)
- Weighted Logistic Regression
- Random Forest
- XGBoost
- XGBoost with SMOTE (Final Model)

Among these, XGBoost combined with SMOTE demonstrated the best performance in handling class imbalance and capturing complex feature relationships.

5.6 Model Evaluation

Model performance was evaluated using multiple metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

ROC-AUC was given special importance due to class imbalance and its ability to measure model discrimination capability.

5.7 Probability-Based Risk Scoring

Instead of binary predictions, the system generates probability scores using:

$$P(y = 1|X)$$

These probabilities are categorized into risk levels:

- Low Risk: $P < 0.3$
- Medium Risk: $0.3 \leq P \leq 0.7$
- High Risk: $P > 0.7$

This allows more informative and actionable health insights.

5.8 Combined Risk Score

An overall health risk score is computed as:

$$\text{Combined Risk} = \frac{1}{n} \sum_{i=1}^{n} P_i$$

where P_i represents the probability of each disease. This provides a unified measure of an individual's overall health risk.

5.9 Model Deployment Pipeline

A modular prediction pipeline is implemented to enable real-time predictions:

- User inputs health parameters
- Preprocessing and scaling are applied
- Trained models are loaded from storage
- Predictions and probabilities are generated
- Risk levels and combined risk score are computed

Models and scalers are stored separately for each disease to ensure flexibility and scalability.

5.10 Model Interpretability

To enhance transparency, SHAP (SHapley Additive exPlanations) is used:

- Identifies important features influencing predictions
- Provides explanation for individual predictions

This improves trust and usability in healthcare applications.

6 Experimental Results and Analysis

6.1 Evaluation Metrics

The performance of the models was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Among these, ROC-AUC was considered the primary evaluation metric due to the presence of class imbalance and its ability to measure model discrimination capability.

6.2 Model Comparison

Multiple machine learning models were evaluated to identify the best-performing approach. The average ROC-AUC scores for each model are presented in Table 4.

Table 4: Model Comparison based on Average ROC-AUC

Model	**Average ROC-AUC**
XGBoost + SMOTE	0.8442
Logistic Regression	0.8403
Weighted Logistic Regression	0.8402
SMOTE Logistic Regression	0.8399
XGBoost	0.8204
Random Forest	0.7821

From the results, XGBoost combined with SMOTE achieved the highest ROC-AUC score, indicating superior performance in distinguishing between positive and negative cases across multiple diseases.

6.3 Disease-Wise Performance Analysis

The performance of the final model (XGBoost + SMOTE) was analyzed individually for each disease.

- **Diabetes:** The model achieved high recall, successfully identifying most positive cases, but with lower precision, indicating false positives.
- **Heart Disease:** Similar to diabetes, the model prioritized recall, ensuring detection of high-risk individuals at the cost of precision.
- **Stroke:** Due to extreme class imbalance, the model achieved high recall but struggled with precision and overall classification balance.
- **High Blood Pressure:** The model performed significantly better, achieving high precision and recall, indicating balanced and reliable predictions.

6.4 Analysis of Class Imbalance Impact

The application of SMOTE significantly improved recall for minority classes such as stroke and heart disease. However, this also led to an increase in false positives, resulting in lower precision for certain diseases.

This behavior is acceptable in healthcare applications where identifying potential high-risk individuals is more critical than minimizing false alarms.

6.5 Key Observations

- XGBoost outperformed traditional models due to its ability to capture complex non-linear relationships.
- SMOTE played a crucial role in improving minority class detection.
- Logistic Regression performed competitively but lacked flexibility in handling non-linear interactions.
- Random Forest showed comparatively lower performance due to its sensitivity to class imbalance.
- High Blood Pressure was the easiest disease to predict due to relatively balanced data distribution.
- Stroke and Heart Disease were the most challenging due to severe class imbalance.

6.6 Visualization of Results

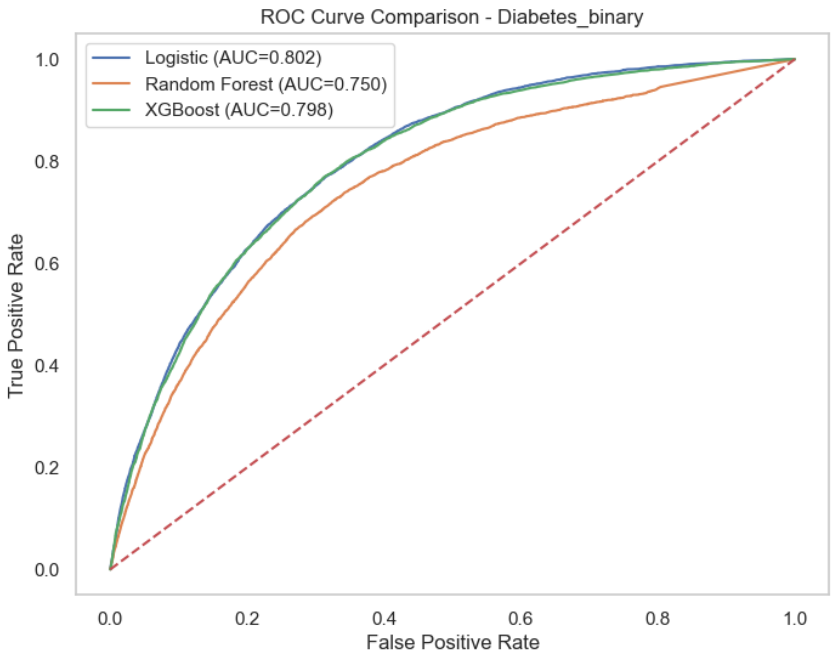

Figure 8: ROC Curve for Model Performance

6.7 Final Model Selection

Based on overall evaluation, XGBoost with SMOTE was selected as the final model due to its:

- Highest ROC-AUC score
- Improved recall for critical diseases
- Ability to handle class imbalance effectively
- Strong generalization performance